Human Computer Interaction and User Interface Design

Detailed Study Notes with Diagrams

Unit 1: Introduction

HCI, GUI, direct manipulation, web interfaces, design principles

Unit 2: Design Process

UCD, human characteristics, interaction speeds, business functions

Includes 16 diagrams, comparison tables, activity guides and review questions

Contents

Contents 2

Unit 1: Introduction 3

1.1 Introduction to Human Computer Interaction (HCI) . . . 3
1.2 Importance of the User Interface . . . 4
1.3 Definition and Importance of Good User Interface Design . . . 4
1.4 Benefits of Good Design . . . 5
1.5 A Brief History of Screen Design . . . 5
1.6 The Graphical User Interface (GUI) . . . 6
1.7 Popularity of Graphics . . . 8
1.8 Concept of Direct Manipulation . . . 8
1.9 Graphical Systems and their Characteristics . . . 9
1.10 Web User Interface . . . 11
1.11 Principles of User Interface Design . . . 13
1.12 Self-Learning Guide (6 hours) . . . 15
1.13 Team Activity Guide (3 hours) . . . 16
1.14 Unit 1 Review Questions . . . 16

Unit 2: Design Process 18

2.1 The Design Process . . . 18
2.2 Human Interaction with Computers . . . 20
2.3 Importance of Human Characteristics . . . 21
2.4 Human Considerations in Interface Design . . . 22
2.5 Human Interaction Speeds . . . 24
2.6 Understanding Business Functions . . . 25
2.7 Self-Learning Guide (6 hours) . . . 27
2.8 Team Activity Guide (3 hours) . . . 27
2.9 Unit 2 Review Questions . . . 28
Quick Revision: Key Terms . . . 28

Unit 1: Introduction

Contact hours: 9 | **Self-learning:** 6 h | **Team activity:** 3 h

> **Unit at a glance**
> Human Computer Interaction (HCI) • importance and definition of a good user interface • benefits of good design • history of screen design • the Graphical User Interface (GUI) and why graphics are popular • direct manipulation • graphical systems and their characteristics • the web user interface • principles of user interface design.

1.1 Introduction to Human Computer Interaction (HCI)

Human Computer Interaction is the discipline concerned with the design, evaluation and implementation of interactive computing systems for human use, and with the study of the major phenomena surrounding them. In simple words, HCI studies how people and computers communicate and how that communication can be made *useful, usable and pleasant*.

Every interactive system contains three ingredients: a **user** (with abilities, limitations, goals), a **computer system** (with functions and data) and an **interface** that lets the two exchange information. All three operate inside a **context of use** - the task being done, the physical environment, the organisation and the culture.

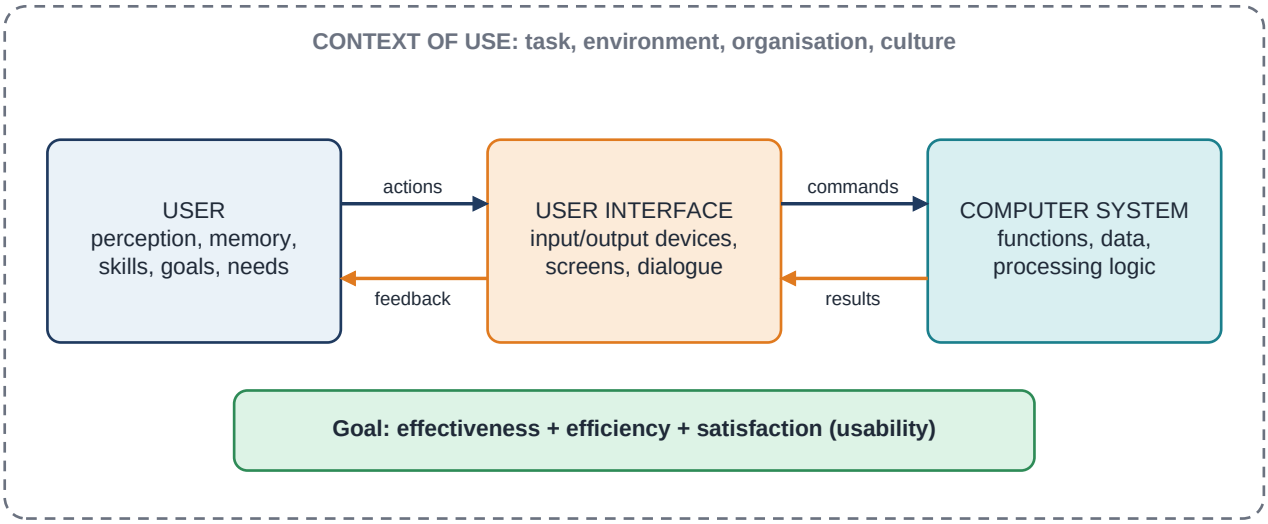


Figure 1.1: The HCI interaction loop - the user acts through the interface and the system responds with feedback.

Why HCI is multidisciplinary

Contributing field	What it adds to HCI
Computer science	Programming, algorithms, software engineering, graphics, architectures for interactive systems.
Cognitive psychology	How people perceive, remember, learn, decide and make errors.
Ergonomics / human factors	Physical fit between people and devices: posture, reach, fatigue, workplace layout.
Graphic and industrial design	Layout, typography, colour, iconography, form of physical devices.
Sociology and anthropology	How groups, organisations and cultures use technology.
Linguistics and communication	Wording of messages, dialogue structure, natural-language and voice interaction.

Goals of HCI

- **Usability goals** - *effectiveness* (does it do what is needed), *efficiency* (with minimum effort), *safety* (protects from dangerous conditions and errors), *utility* (right functions provided), *learnability* (easy to learn) and *memorability* (easy to remember after a break).
- **User experience goals** - the system should also be satisfying, enjoyable, engaging, motivating, aesthetically pleasing, rewarding and emotionally fulfilling.
- **Business goals** - lower training and support cost, fewer errors, higher productivity, better customer retention.

> **Usability (ISO 9241-11)**
> The extent to which a product can be used by specified users to achieve specified goals with **effectiveness, efficiency and satisfaction** in a specified context of use.

1.2 Importance of the User Interface

To an ordinary user the **interface *is* the system**: the user never sees the database, the algorithms or the network - only what is displayed and how it responds. A brilliant back end hidden behind a confusing interface will be judged a poor product. The interface therefore determines whether people can use the power that has been built into the software.

- **It is the point of contact** between human and machine; all input and output passes through it.
- **It shapes perception of quality** - users equate a clean, responsive interface with a reliable system.
- **It drives adoption** - people abandon apps and websites they find confusing within seconds; competitors are one click away.
- **It affects safety** - in control rooms, medical devices, aircraft cockpits and vehicles, a misleading display can cause accidents. Investigations of several industrial and aviation incidents have cited poor display and control design as a contributing factor.
- **It affects cost** - a large share of software effort, and of support calls, relates to the user interface.

Symptoms of a poor interface

- Users cannot find a function that exists; they need a manual or training for basic tasks.
- Frequent errors, wrong clicks, data-entry mistakes, and no easy way to recover.
- Long task times, frustration, low morale, and high staff turnover in workplace systems.
- Many help-desk calls and workaround habits (sticky notes, paper forms).

1.3 Definition and Importance of Good User Interface Design

The **user interface** is the part of a computer and its software that people can see, hear, touch, talk to, or otherwise understand or direct. It has two main components: the **input mechanism** (keyboard, mouse, touch, voice, camera, sensors) and the **output mechanism** (screen, sound, haptics, printer). The interface also includes the **dialogue** - the structured conversation between user and computer - and its **documentation and help**.

> **Good user interface design**
> Design that makes the interface **effective, efficient, easy to learn and remember, error-tolerant and satisfying** for its intended users, by matching the interface to human capabilities and the task rather than forcing people to adapt to the machine.

Characteristics of a good interface

Characteristic	Meaning	Example
Simple	Only necessary elements; no clutter	Search box on a search-engine home page
Natural / familiar	Uses concepts the user already knows	Trash / recycle-bin icon for deleting files
Consistent	Same action, same result everywhere	Ctrl+C always copies
Responsive	Immediate, visible feedback	Button changes state when pressed
Forgiving	Easy to undo; prevents disasters	Undo, confirmation before permanent delete
Accessible	Usable by people with different abilities	Alt text, keyboard navigation, sufficient contrast

Importance can be summed up in one sentence: interface quality decides whether a capable system becomes a *successful* one. It influences user productivity, error rates, training needs, safety, customer loyalty and, ultimately, the commercial success of the product.

1.4 Benefits of Good Design

Investing in interface design early is far cheaper than repairing usability problems after release. Benefits accrue to users, to the organisation and to the development team.

Beneficiary	Benefits
Users	Faster task completion; fewer errors; lower mental effort and stress; higher satisfaction; confidence to explore features.
Organisation / customer	Reduced training time and cost; fewer help-desk calls; higher staff productivity; improved customer retention and sales; better brand image; lower risk of costly errors.
Developers	Fewer late design changes and rework; clearer requirements; less maintenance; reduced support burden.
Society	Systems usable by more people, including older adults and people with disabilities (inclusion, equality of access).

Cost of change
The later a usability problem is discovered (concept, prototype, coding, release), the more expensive it is to fix. Usability testing on cheap paper prototypes costs a fraction of re-engineering a shipped product.

1.5 A Brief History of Screen Design

Interface design evolved as hardware evolved. Each generation removed some burden from the user and moved it to the machine.

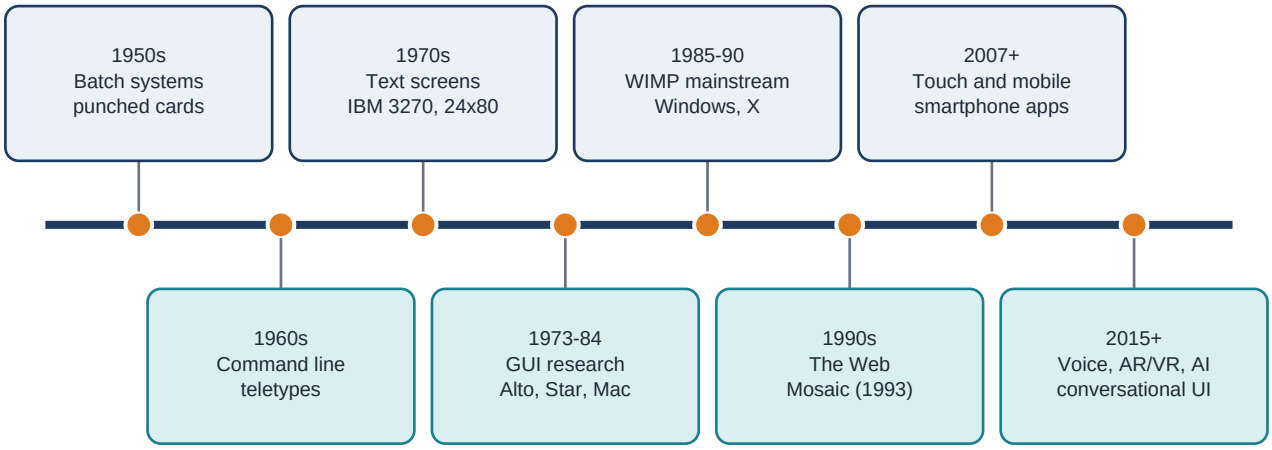


Figure 1.2: Timeline of user interface evolution, from punched cards to conversational and immersive interfaces.

Stages in detail

- **Batch processing (1950s-60s).** Jobs were prepared offline on punched cards or tape; results arrived hours later as printouts. There was no real-time interaction - a single typing error meant waiting for the next run.
- **Command-line interfaces (1960s-70s).** Teletypes and video terminals allowed conversation in text. Users typed cryptic commands (for example UNIX shells). Powerful for experts but demanding on memory and syntax.
- **Text-based full-screen displays (1970s-80s).** Mainframe terminals such as the IBM 3270 showed 24 lines x 80 columns of characters. Screens were often *dense, cryptic and cluttered*, with data entry through form-like fields. Researchers began studying screen density, grouping and alignment, giving rise to screen-design guidelines.
- **GUI research (1973-84).** Xerox PARC built the Alto (1973) and later the Star (1981) with windows, icons, a mouse and the desktop metaphor. Apple Lisa (1983) and Macintosh (1984) brought the ideas to a wider market.
- **WIMP goes mainstream (mid-1980s onward).** Microsoft Windows, the X Window System and OS/2 standardised windows, icons, menus and pointers, together with published style guides that promoted consistency.
- **The Web (1990s).** Hypertext and browsers (Mosaic 1993) created a new kind of interface centred on *content and navigation*, available on any platform.
- **Mobile and touch (2007+).** Smartphones and tablets introduced multitouch, gestures, sensors and small screens; responsive and app-based design became essential.
- **Post-WIMP (2015+).** Voice assistants, chatbots, wearables, augmented and virtual reality and AI-assisted interfaces extend interaction beyond screen, keyboard and mouse.

1.6 The Graphical User Interface (GUI)

A **Graphical User Interface** lets users interact with a computer through visual objects - windows, icons, menus, buttons and pointers - manipulated with a pointing device or touch, instead of typing commands. The classic GUI is described by the acronym **WIMP**: **W**indows, **I**cons, **M**enus, **P**ointers.

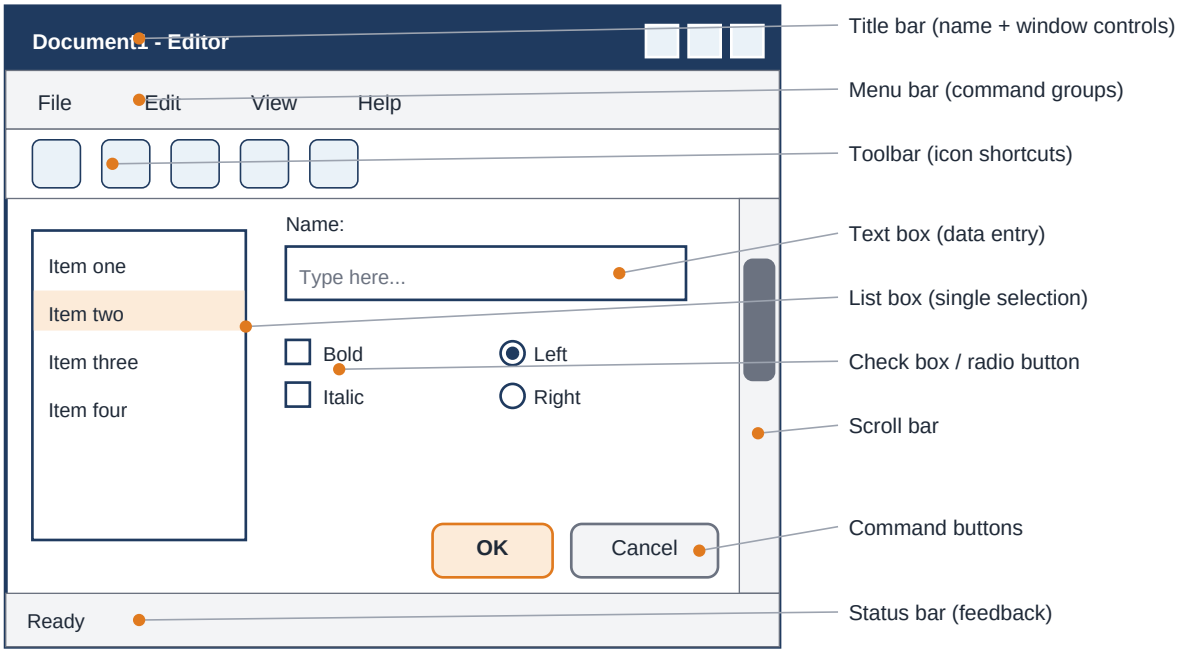


Figure 1.3: Anatomy of a typical GUI window and its common controls.

Main GUI components

Component	Purpose	Notes
Window	Separate viewing area for an application or document	Has title bar, border, resize and scroll controls
Icon	Small picture representing an object or action	Must be recognisable; often paired with a text label
Menu	List of commands grouped logically	Menu bar, pull-down, cascading, pop-up/context menus
Pointer / cursor	Shows position controlled by the input device	Changes shape to show state (arrow, hand, I-beam, busy)
Controls (widgets)	Buttons, text boxes, lists, sliders, check boxes	Standard behaviour makes them predictable
Dialog box	Short, focused exchange for input or confirmation	Modal (blocks) or modeless (does not block)

Advantages and disadvantages of GUIs

Advantages

- Easy to learn and remember (recognition instead of recall)
- Consistent look and feel across applications
- Multiple tasks visible at once
- Immediate visual feedback
- Suits novices and experts (shortcuts)

Disadvantages

- Needs more memory, processing power and screen space
- Icons can be ambiguous
- Pointing is slower than typing for some experts
- Poor design can still produce clutter
- Harder to script/automate than commands

1.7 Popularity of Graphics

Graphical interfaces replaced text screens because the human brain processes pictures and spatial arrangements very effectively. The main reasons for the popularity of graphics are:

- **Symbols recognised faster than text.** A familiar picture (printer, folder, magnifier) is understood at a glance.
- **Faster learning and retention.** People remember images better than words; visual cues reduce memory load.
- **Familiar metaphors.** The desktop, folder, trash can and shopping cart map computer actions onto real-world knowledge.
- **Information compression.** A chart or map conveys relationships that would need pages of numbers or text.
- **Spatial organisation.** Position, size, grouping and colour show structure, priority and relationship instantly.
- **What You See Is What You Get (WYSIWYG).** The screen shows the final result of editing, reducing surprise.
- **Recognition over recall.** Users pick visible options rather than remembering command names.
- **Attractiveness and engagement.** Well-designed visuals are more pleasant and increase trust.

> **Caution**
> Graphics are not automatically better. Poorly chosen icons, decorative clutter, excessive colour and heavy images can slow the system, distract users and reduce usability. Graphics must **serve the task**.

1.8 Concept of Direct Manipulation

Direct manipulation, a term introduced by Ben Shneiderman in 1983, is an interaction style in which users act directly on visible representations of objects of interest, using physical actions such as pointing, dragging and resizing, and see the results immediately. Instead of typing "move file A to folder B", the user simply *drags the file icon onto the folder icon*.

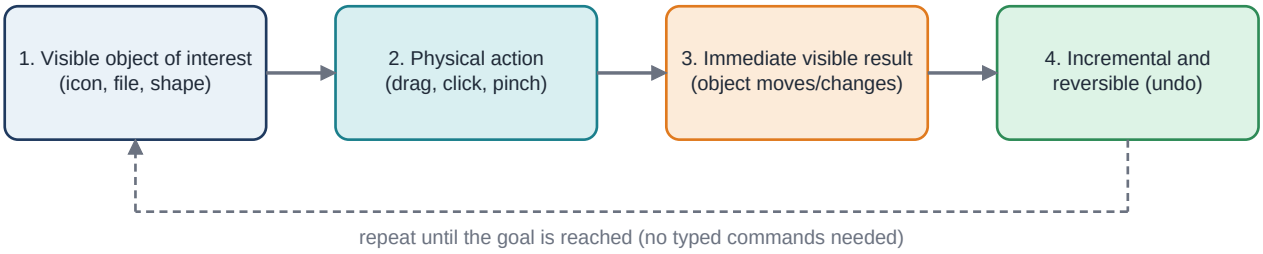


Figure 1.4: The direct-manipulation cycle.

Core properties

- **Continuous visibility** of the objects and actions of interest.
- **Physical actions** (click, drag, pinch) replace complex command syntax.
- **Rapid, incremental and reversible** actions, with results that are immediately visible.
- **Layered learning** - novices succeed with a few simple actions; experts add shortcuts and advanced features.
- Users feel **in control** and directly engaged with the task rather than communicating through an intermediary.

Examples

Dragging a file to the trash, resizing a window by its border, moving a slider to change volume, pinch-to-zoom on a phone map, drawing a shape in a graphics editor, re-ordering slides by dragging thumbnails, and swiping a list item to delete it.

Advantages and limitations

Advantages	Limitations
Easy to learn; novices become productive quickly	Not all tasks have a natural visual representation (e.g. "replace all X with Y")
Fewer errors and easy recovery through undo	Precision pointing can be tiring (RSI, small touch targets)
Encourages exploration; high satisfaction	Repetitive or batch operations are slower than commands or scripts
Reduces need for memorising syntax	Needs more screen space and graphics capability

Gulf of execution and gulf of evaluation (Norman)

Donald Norman explains interaction as a cycle of seven stages. Two "gulfs" can separate the user from the system. The **gulf of execution** is the gap between what the user wants to do and the actions the system provides. The **gulf of evaluation** is the gap between the system state and the user's ability to perceive and interpret it. Direct manipulation narrows both gulfs because the interface object *is* the thing being controlled, and the effect of every action is visible.

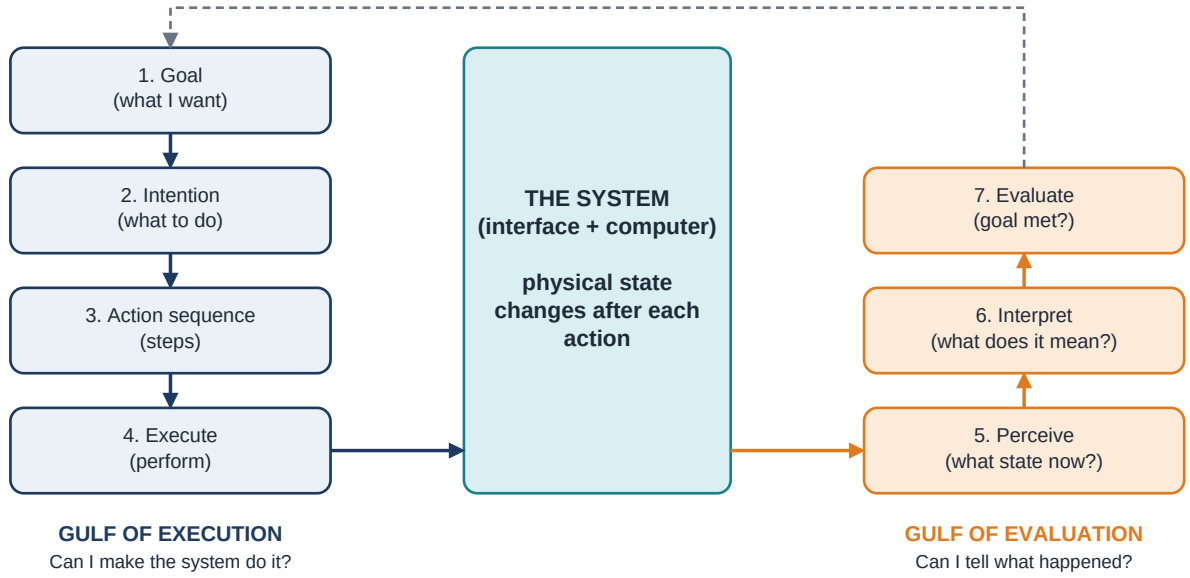


Figure 1.5: Norman's seven-stage action cycle and the two gulfs.

1.9 Graphical Systems and their Characteristics

A graphical system is an operating environment (Windows, macOS, Linux desktops, Android, iOS) that provides windows, menus and controls through a standard toolkit. Its characteristics can be divided into **presentation** (what is seen) and **interaction** (how it is used).

Characteristics of graphical systems

Characteristic	Description
Sophisticated visual presentation	Proportional fonts, colour, icons, images, animation, 3-D effects to convey meaning and hierarchy.
Restricted set of interface options	A limited, standard collection of controls and window types keeps the system understandable.
Visibility of objects and actions	Available objects, commands and their states are shown, so users recognise rather than recall.
Consistency	Common behaviour, position and appearance of controls across applications (style guides).
Ease of learning and use	Standard metaphors and transferable skills reduce training.
Immediate feedback	Every action produces a visible response - highlight, sound, progress, message.
Reversibility and error prevention	Undo/redo, confirmation and disabled invalid options protect the user.
Direct manipulation	Objects are selected and acted on with a pointer or touch.
Multiple views and multitasking	Several windows or applications can be open and switched between simultaneously.
Device independence	Applications talk to standard drivers, not to specific devices.

Windows

A window is a rectangular area of the screen containing a document, application or message. Typical elements: title bar, borders, system/control menu, minimise-maximise-close buttons, scroll bars, status bar and a resizing handle. Windows are classified as:

Window type	Use
Primary window	Main application window; presents the principal objects and tasks.
Secondary window	Supports the primary window: dialog boxes, property sheets, palettes, pop-ups.
Dialog box	Requests or provides information; may be modal (must be dismissed first) or modeless.
Message box	Shows information, warnings, errors or asks a question (OK / Yes / No).
Palette / toolbox	Persistent set of tools or options, e.g. drawing tools.

Menus

- **Menu bar** - permanent top-level categories (File, Edit, View...).
- **Pull-down and cascading menus** - lists that open from a menu-bar item; cascades show submenus.
- **Pop-up / context menus** - appear near the pointer with commands relevant to the selected object.
- **Guidelines** - group related items, use meaningful names, order by frequency or logic, keep length manageable, show keyboard shortcuts, and grey out (rather than hide) unavailable options.

Controls (widgets) and when to use them

Control	Use for
Command button	Immediate actions (Save, Cancel, Submit).
Radio buttons	Choose exactly **one** from a small, visible set (2-6 items).
Check boxes	Choose **zero or more** independent options.
List box	Select from a longer list, with items visible.
Drop-down / combo box	Select from a longer list in a compact space; combo also allows typing.
Text box / text area	Free-form entry of short or long text.
Spin box / slider	Numeric or range values where approximate or stepped adjustment is useful.
Toggle switch	Turn a single setting on or off with immediate effect.
Tabs	Organise related pages of settings within one window.

Icons, pointers and pointing devices

Icons are compact visual representations of objects or actions; they should be simple, distinct, familiar and usually labelled. **Pointer shapes** communicate state - arrow (select), I-beam (text), hand (link), hourglass/spinner (busy), resize arrows. **Pointing devices** include mouse, trackpad, touch screen, stylus, joystick and trackball; each has different speed, precision and fatigue characteristics.

1.10 Web User Interface

A **web user interface** is presented through a browser and delivered over the Internet. The user navigates between *pages* using hyperlinks, fills in forms and receives content that may be static or dynamically generated. The web has become the dominant platform for information, commerce, education, banking and entertainment.

Popularity of web interfaces

- **Universal access** - any device with a browser and connection; no installation.
- **Platform independence** - one application works on Windows, macOS, Linux, Android and iOS.
- **Hypertext navigation** - users move freely by following links; easy to learn.
- **Easy publishing and updating** - changes reach all users instantly; no distribution cost.
- **Rich media and interactivity** - text, images, audio, video, and application-like behaviour.
- **Global reach and e-commerce** - organisations can serve customers around the clock.

Characteristics of web interfaces

- **Content-centred** - purpose is usually to find, read or transact, rather than to operate a complex tool.
- **Page-based and link-driven** - navigation is central; the user may enter at any page (search engines, bookmarks).
- **Uncontrolled environment** - screen size, browser, plug-ins, network speed and user skills all vary.
- **Latency** - download and server delays affect experience; page weight matters.
- **Less control by designer** - users can resize windows, change fonts, block scripts.
- **Responsive design** - layout adapts to desktop, tablet and phone.
- **Open, competitive, and public** - alternatives are always a click away, so first impressions and speed are critical.

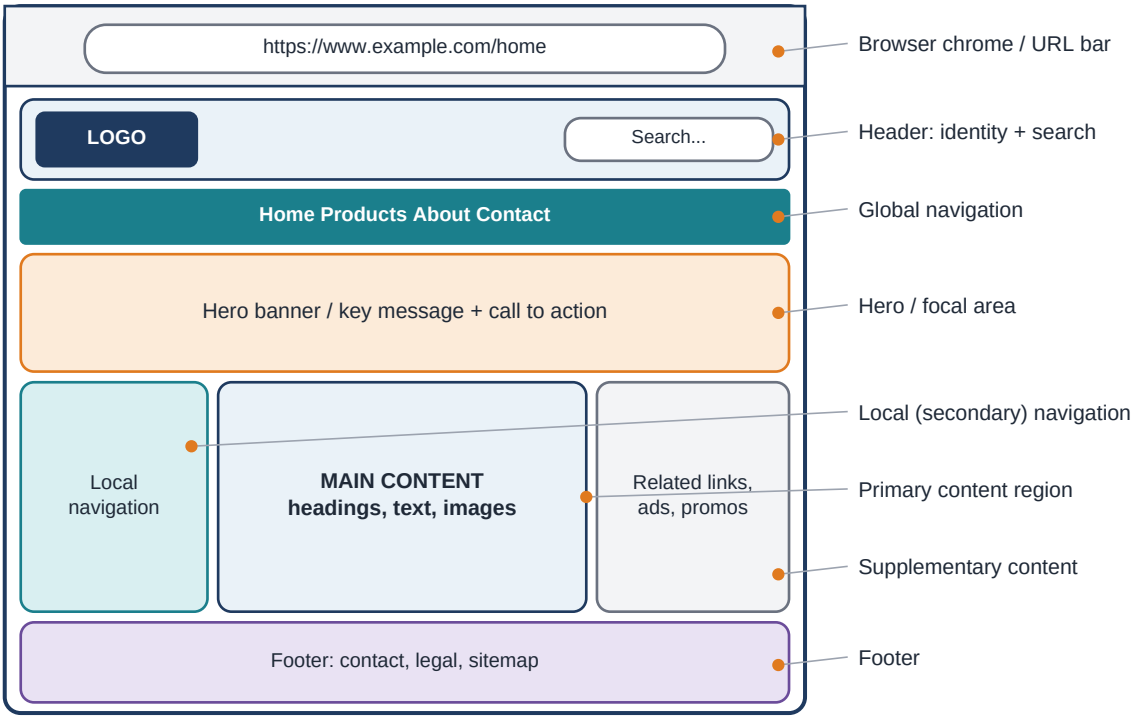


Figure 1.6: Typical regions of a web page.

Web interface versus GUI

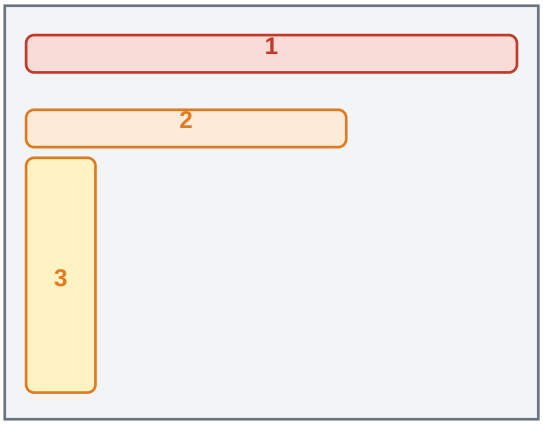
Aspect	Graphical (desktop) user interface	Web user interface
Primary focus	Data manipulation and application tasks	Content, information and navigation
Unit of interaction	Screens / windows	Pages linked by hyperlinks
Style control	Designer controls appearance closely	Shared with browser and user (fonts, window size)
Navigation	Menus, dialogs, task flow	Hyperlinks, back button, search, site map
Interaction	Direct manipulation, rich, immediate	Click-and-wait; improved by scripting and AJAX
Performance	Local; fast and predictable	Depends on network and server
Users	Often trained, known audience	Anyone; wide range of skills
Installation and updates	Installed per machine	None; updated centrally

Site structure and reading behaviour



Figure 1.7: Common site structures.

Users scan web pages rather than read them word by word. Eye-tracking studies commonly show an **F-shaped** scanning pattern on text-heavy pages. Place the most important information at the top and left, use meaningful headings, short paragraphs and bullet lists.



1. First horizontal sweep across the top
2. Shorter second sweep lower down
3. Vertical scan down the left edge

Design implication: put key messages, links and calls to action where the eyes land - top and left first.

Figure 1.8: F-shaped reading pattern.

Web design guidelines

- Make the purpose of the site and the home page obvious within seconds; give a clear site identity on every page.
- Provide consistent, visible navigation with meaningful link labels, breadcrumbs and a search facility.
- Keep pages light for fast loading; optimise images; give feedback for long operations.
- Use plain language, headings and white space; limit the number of fonts and colours.
- Design for accessibility (alt text, contrast, keyboard focus, resizable text) and for all screen sizes.
- Make forms short, validate input near the field and provide clear, polite error messages.

1.11 Principles of User Interface Design

Design principles are broad, well-tested rules that guide decisions when no specific guideline applies. They come from research in psychology, ergonomics and decades of practice. Several authoritative sets exist; they overlap considerably.

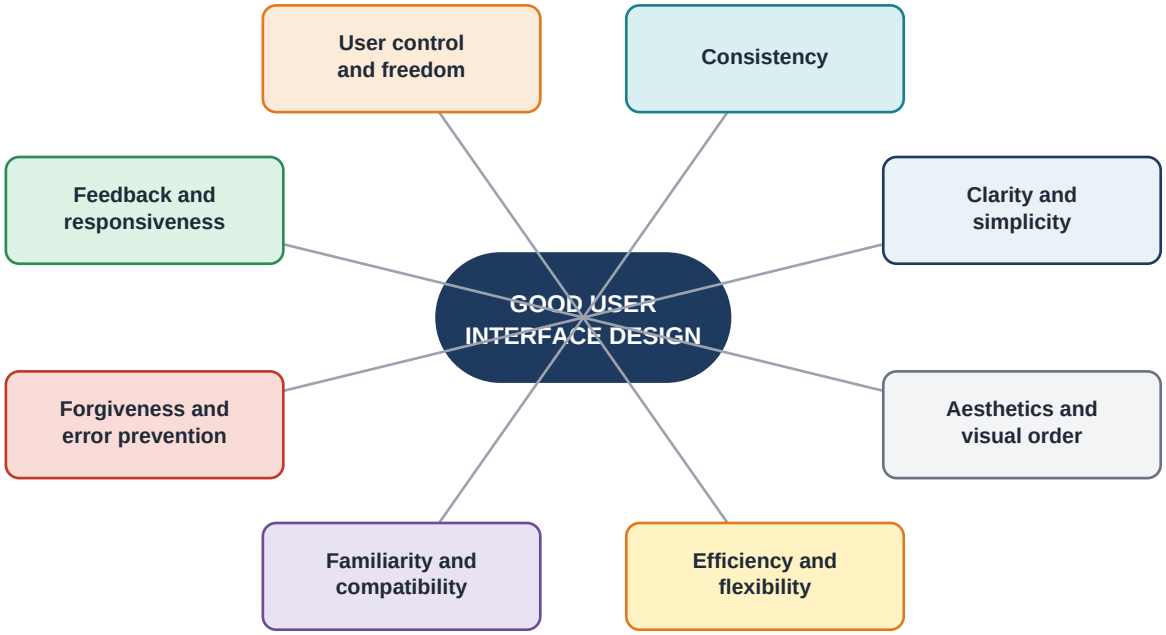


Figure 1.9: Major families of user interface design principles.

A. General principles (Galitz-style)

Principle	Meaning
Aesthetically pleasing	Provide visual appeal: balance, clear grouping, restrained colour and typography.
Clarity	Visual, conceptual and linguistic clarity - meaning is obvious and unambiguous.
Compatibility	Match the user's expectations, tasks, and other systems they use.
Comprehensibility	The user knows what to do, when, and why; structure is understandable.
Configurability	Allow personalisation (preferences, layout) where useful.
Consistency	Same terminology, layout and behaviour for same situations.
Control	The user, not the system, is in control of the interaction and can interrupt or cancel.
Directness	Provide direct ways to accomplish tasks; make alternatives visible.
Efficiency	Minimise eye and hand movement, steps and time; provide shortcuts.
Familiarity	Use concepts, words and images from the user's world.
Flexibility	Support different skills, styles and preferences.
Forgiveness	Tolerate errors; prevent serious ones; allow undo.
Predictability	The user can anticipate what will happen next.
Recovery	Help users recover from mistakes with clear messages and remedies.
Responsiveness	Acknowledge each action promptly; keep the user informed.
Simplicity	Present only what is needed; break complex tasks into manageable steps.
Transparency	Let the user focus on the task, not on the technology.

B. Nielsen's ten usability heuristics

#	Heuristic	Example
1	Visibility of system status	Progress bar during upload
2	Match between system and real world	Shopping cart, trash can
3	User control and freedom	Undo, cancel, back
4	Consistency and standards	Standard icons and platform conventions
5	Error prevention	Disable invalid options; confirm destructive actions
6	Recognition rather than recall	Recent files, visible menus
7	Flexibility and efficiency of use	Keyboard shortcuts for experts
8	Aesthetic and minimalist design	No irrelevant information
9	Help users recognise, diagnose and recover from errors	Plain-language error with a suggested fix
10	Help and documentation	Searchable, task-oriented help

C. Norman's design principles

- **Visibility** - relevant controls and state are visible.
- **Feedback** - each action gets a clear response.
- **Constraints** - limit possible actions to prevent errors.
- **Mapping** - relationship between control and effect is natural (stove knobs matching burners).
- **Consistency** - similar operations work similarly.
- **Affordance** - an object's appearance suggests how it is used (a button looks pressable).

D. Shneiderman's eight golden rules

1) Strive for consistency. 2) Enable frequent users to use shortcuts. 3) Offer informative feedback. 4) Design dialogues that yield closure. 5) Offer simple error handling. 6) Permit easy reversal of actions. 7) Support internal locus of control. 8) Reduce short-term memory load.

> **Trade-offs**
> Principles sometimes conflict: simplicity versus flexibility, consistency versus innovation, efficiency versus forgiveness. Good designers weigh the trade-offs using knowledge of the users, tasks and context - and confirm with testing.

1.12 Self-Learning Guide (6 hours)

Use the following structured plan to cover the self-learning component.

Task	What to do	Output
1. Evolution of interfaces	Read about command line, text screens, GUI, web, mobile. Try a terminal command, then do the same task in a GUI.	One-page timeline with 2 screenshots per era
2. Study interface examples	Explore a graphical application (word processor, image editor) and a web application (e-commerce, banking, e-learning).	List of controls used and why
3. Identify principles	For each principle in 1.11 find one good and one bad example in software you use.	Table: principle - example - verdict
4. Evaluate usability	Perform a heuristic evaluation (Nielsen) of an app you know: rate each heuristic 0-4, note severity.	Evaluation sheet with 5 problems and fixes

Quick heuristic evaluation sheet

Heuristic	Problem found	Severity (0-4)	Suggested fix
Visibility of system status			
Error prevention			
Consistency and standards			
Recognition rather than recall			

Severity scale: 0 = not a problem, 1 = cosmetic, 2 = minor, 3 = major, 4 = catastrophic.

1.13 Team Activity Guide (3 hours)

Activity: compare the GUIs of different desktop and web applications, identify good and poor practices against established principles, and suggest improvements.

- **Step 1 (20 min) - Choose.** Pick two comparable products, e.g. a desktop word processor and its web version, or two shopping sites.
- **Step 2 (45 min) - Perform tasks.** Each member completes the same 3-5 tasks (create account, search, edit, checkout). Record time, clicks and errors.
- **Step 3 (45 min) - Analyse.** Fill in the comparison matrix below; tag each observation with the principle it supports or violates.
- **Step 4 (30 min) - Improve.** Sketch (paper or wireframe) two or three improvements for the worst issues.
- **Step 5 (40 min) - Present.** 8-10 minutes: context, method, findings, before/after sketches, lessons.

Comparison matrix template

Criterion	Application A	Application B	Principle involved
Layout and visual hierarchy			Clarity, simplicity
Navigation and menus			Consistency, control
Feedback and status			Responsiveness
Error handling and undo			Forgiveness, recovery
Icons and labels			Familiarity, clarity
Accessibility (contrast, keyboard)			Flexibility, inclusion
Task time / clicks			Efficiency

Worked mini-example

A government form website requires date entry in a plain text box with an unspecified format and, on error, clears the whole form. **Violations:** error prevention, forgiveness, clarity. **Improvement:** use a date picker or a hint such as DD/MM/YYYY, validate on the field, keep entered data and show a specific error message beside the field.

1.14 Unit 1 Review Questions

- Define HCI. Explain its goals and the disciplines that contribute to it.
- Why is the user interface important? Discuss the benefits of good design.
- Trace the history of screen design with the significant milestones.
- What is a GUI? Describe its components and advantages/disadvantages.
- Why are graphics popular in user interfaces? Are there any drawbacks?

- Explain direct manipulation with examples. What are its limitations?
- List the characteristics of graphical systems. Describe window types and menu types.
- Compare web interfaces with GUIs. State the characteristics and reasons for popularity of web interfaces.
- Explain any ten principles of user interface design with examples.
- Explain Norman's gulfs of execution and evaluation.

Unit 2: Design Process

Contact hours: 8 | **Self-learning:** 6 h | **Team activity:** 3 h

> **Unit at a glance**
> The interface design process • obstacles and common usability problems • human interaction with computers • importance of human characteristics • human considerations in interface design • human interaction speeds • understanding business functions (requirements analysis, standards, training and documentation).

2.1 The Design Process

Designing an interface is not a single act of drawing screens. It is a **structured, iterative process** in which designers learn about users and their tasks, propose solutions, build prototypes, test them with real users, and refine them. This philosophy is called **user-centred design (UCD)** or human-centred design.

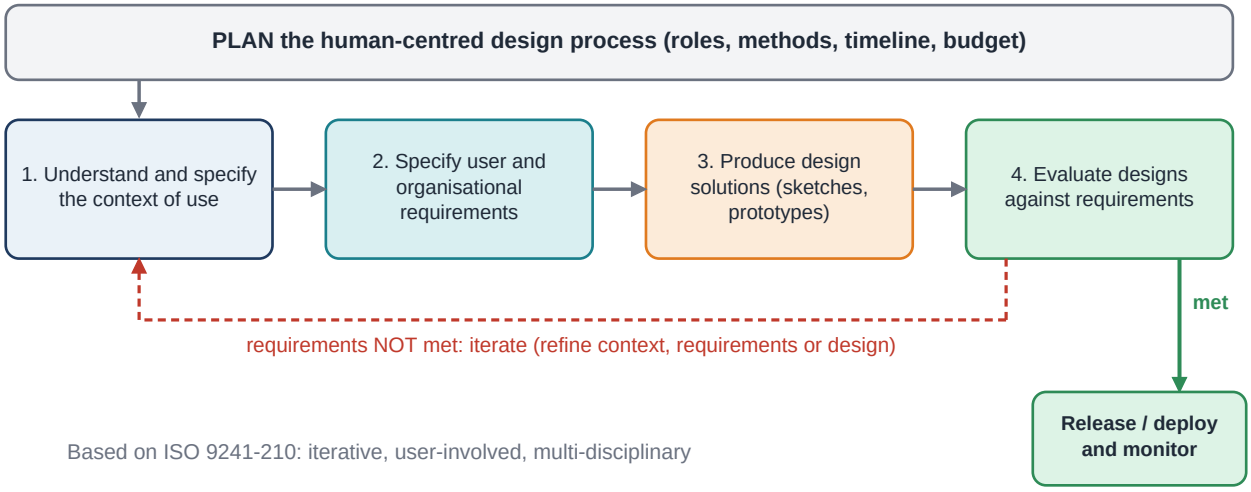


Figure 2.1: The user-centred design cycle (after ISO 9241-210).

Principles of user-centred design

- Design is based on an explicit understanding of users, tasks and environments.
- Users are involved throughout design and development.
- Design is driven and refined by user-centred **evaluation**.
- The process is **iterative**.
- Design addresses the whole user experience.
- The design team includes multidisciplinary skills and perspectives.

Obstacles and pitfalls in the development process

Even with good intentions, projects run into recurring problems that produce poor interfaces:

- **No early usability involvement** - the interface is treated as a finishing touch after the software architecture is fixed.
- **Ignorance of users** - designers rely on assumptions ("I am like the user") rather than study.
- **Lack of design guidelines, standards and expertise** in the team.
- **Poor communication** between developers, designers, users and management.
- **Feature creep** and unclear, changing requirements.

- **Pressure of time and cost** leading to skipped testing.
- **Reluctance to change** when problems are exposed late.

Common usability problems

- Ambiguous menus and icons; unclear labels.
- Unclear or missing feedback about what the system is doing.
- Inconsistent conventions between screens or applications.
- Overly complex navigation; too many steps or too much on one screen.
- Poor error messages that blame the user or give no remedy.
- Excessive demands on memory (codes, remembering data across screens).
- Poor layout: cluttered, poorly aligned or inappropriately coloured screens.
- Slow or unpredictable system response.
- Inaccessible design for older users or those with disabilities.

Design tips

- Involve users early; observe them working.
- Follow guidelines and platform style guides.
- Prototype quickly and cheaply, test, and revise.
- Build a design team with complementary skills and clear roles.
- Document decisions and define measurable usability targets.
- Test and retest; do not rely on opinion.

Steps in the interface design process

A widely taught structured sequence expresses the process as a series of steps, each producing decisions that feed the next:

Step	Activity
1	Know your user or client - identify characteristics, tasks, environment.
2	Understand the business function - goals, requirements, workflows.
3	Understand the principles of good screen and interface design.
4	Develop system menus and navigation structure.
5	Select the proper kinds of windows.
6	Select the proper device-based controls (mouse, keyboard, touch...).
7	Choose the proper screen-based controls (buttons, lists, fields...).
8	Write clear text and messages.
9	Provide effective feedback, guidance and assistance.
10	Provide effective internationalisation and accessibility.
11	Create meaningful graphics, icons and images.
12	Choose proper colours.
13	Organise and lay out windows and pages.
14	Test, test, and retest.

Prototyping and evaluation

Fidelity	Form	Purpose
Low	Paper sketches, storyboards, cardboard mock-ups	Explore concepts quickly and cheaply; anyone can contribute
Medium	Wireframes, clickable mock-ups	Test navigation and layout
High	Interactive prototype or working code	Test detailed interaction, visual design and timing

Evaluation methods: usability testing with representative users (think-aloud), heuristic evaluation by experts, cognitive walkthrough, questionnaires (for example SUS - System Usability Scale), A/B testing, and analysis of logs and support requests.

2.2 Human Interaction with Computers

Humans are information processors: they take in information through the senses, process it in the brain and produce a response through movement or speech. Understanding this chain lets designers predict what people can and cannot do at the interface. The best-known simplified description is the **Model Human Processor** of Card, Moran and Newell.

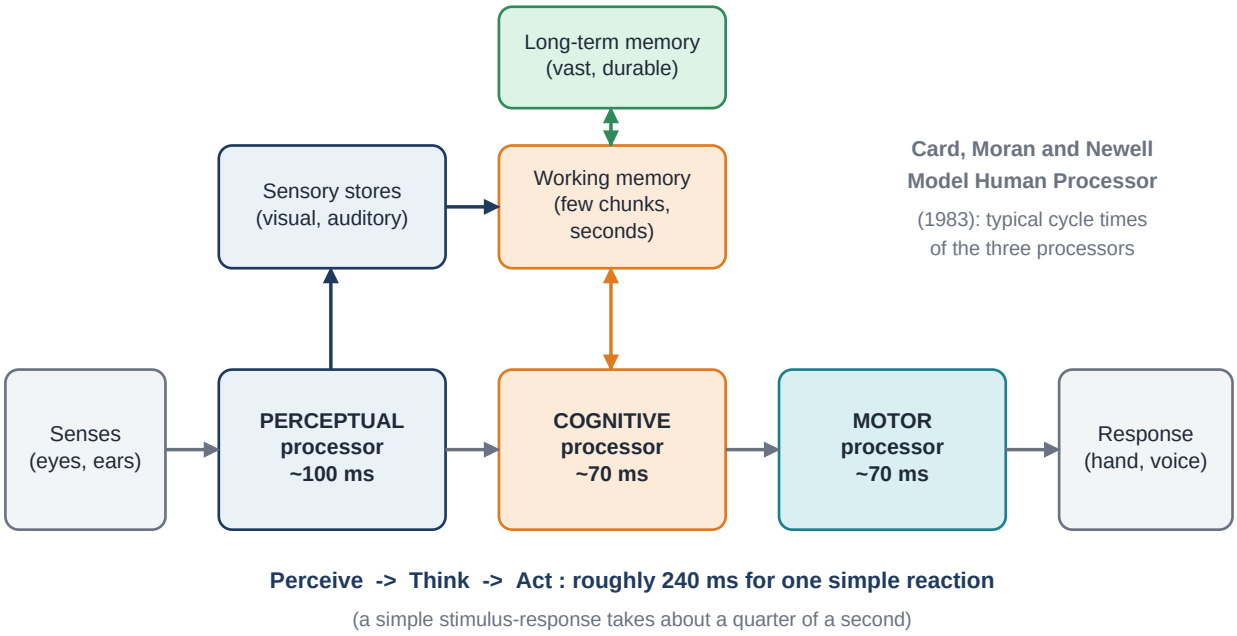


Figure 2.2: Model Human Processor - perceptual, cognitive and motor subsystems with memory stores.

The three subsystems

- **Perceptual system** - senses detect stimuli (mainly vision and hearing) and pass them to sensory stores.
- **Cognitive system** - uses working memory and long-term memory to interpret, decide and plan.
- **Motor system** - converts decisions into actions such as pressing keys, moving the mouse or speaking.

Memory

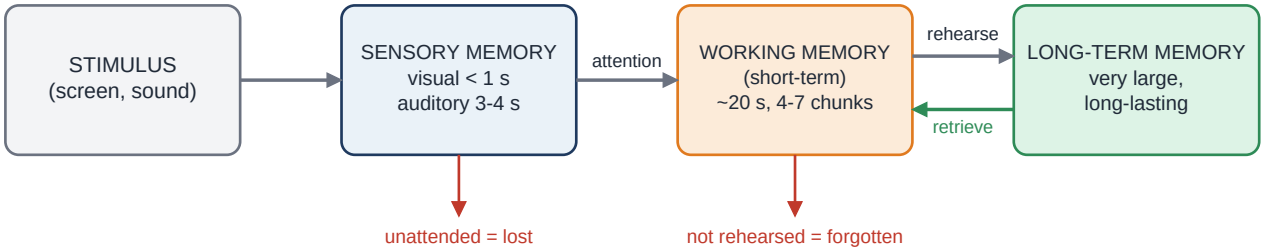


Figure 2.3: Multi-store memory model and its consequences for interface design.

Memory	Characteristics	Design implication
Sensory memory	Very brief (visual under a second, auditory a few seconds); large capacity	Information must be noticed quickly; flashing or transient messages are missed
Working (short-term) memory	Limited capacity - roughly 7 +/- 2 items (Miller); more recent work suggests about 4 chunks; lasts seconds unless rehearsed	Do not require users to remember information across screens; chunk data (phone numbers, codes); show context
Long-term memory	Practically unlimited and durable; retrieval may be difficult; strengthened by meaning and repetition	Use familiar concepts, metaphors and consistent patterns; support **recognition** instead of recall

Recognition versus recall

Recognising an option in a menu is easy; recalling a command name from memory is hard. GUIs rely on recognition; command lines rely on recall. This single difference explains much of the GUI's ease of learning.

Mental models

A **mental model** is the user's internal picture of how a system works, built from experience and metaphors. If the designer's *conceptual model* is presented clearly by the interface (the *system image*), user models will be accurate and behaviour predictable. Mismatches produce errors and surprise - for example users who think a "folder" is a physical container may be confused when the same file appears in two places through a shortcut.

2.3 Importance of Human Characteristics

People differ in physical and mental abilities, but share general characteristics that determine how well they can use an interface. Ignoring these characteristics leads to systems that are slow, tiring and error-prone. Important characteristics include:

Characteristic	Explanation	Design consequence
Perception	How the brain organises sensory input into meaningful patterns (proximity, similarity, closure, continuity)	Group related items, use alignment and white space to show structure
Visual acuity	Detail can be seen sharply only in a small central area (fovea, roughly 2 degrees); the periphery detects movement and gross shape	Keep related information close; put key items in the focus area; use motion sparingly to attract attention
Colour vision	About 8% of men and 0.5% of women have some colour-vision deficiency; sensitivity varies with brightness and age	Never use colour as the only cue; ensure contrast; use redundant coding (shape, text)
Attention	Limited and selective; easily captured by movement and change	Avoid distracting animation; highlight only what matters

Characteristic	Explanation	Design consequence
Learning	Occurs through repetition, meaning and feedback; transfers from similar systems	Use consistent conventions; provide guidance for first-time use
Skill	Practice reduces conscious effort; experts work in chunks and want speed	Offer shortcuts, defaults, type-ahead
Reading and scanning	Reading from screens is slower and more tiring than from paper; users scan	Short lines, headings, plain language, adequate font size and contrast
Movement control	Precision, speed and fatigue limits (see Fitts's law)	Large, well-spaced targets; minimise repetitive movement
Individual differences	Age, culture, language, experience, disabilities, preferences	Offer flexibility, configurability and accessibility features

Gestalt principles of visual perception

The Gestalt psychologists showed that people perceive wholes before parts. Designers use these principles to communicate grouping without borders:

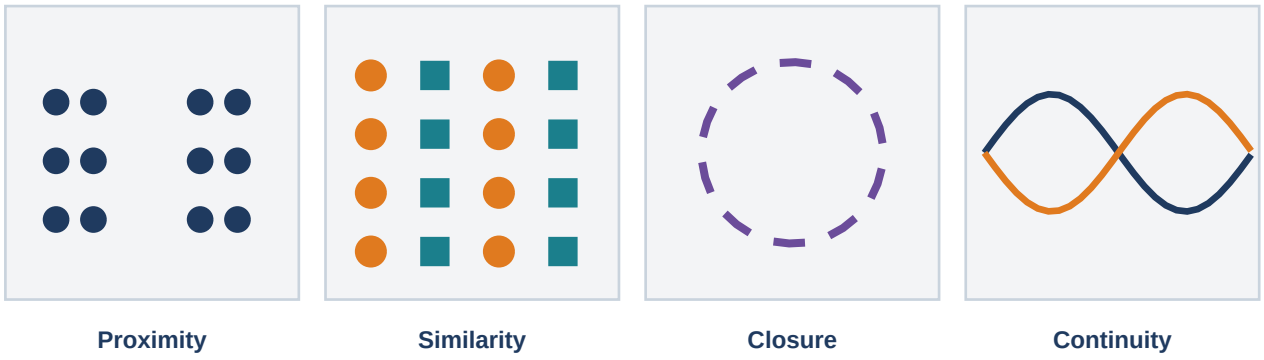


Figure 2.4: Four Gestalt principles - proximity, similarity, closure and continuity.

- **Proximity** - items close together are seen as a group (labels near their fields).
- **Similarity** - items that look alike are seen as related (same style for all links).
- **Closure** - the mind completes incomplete shapes (icons need not be fully drawn).
- **Continuity** - the eye follows smooth paths (aligned edges guide scanning).
- Also **figure/ground**, **common region** (a box makes a group) and **symmetry**.

2.4 Human Considerations in Interface Design

The designer must translate knowledge of people into concrete interface decisions. The main considerations are grouped below.

Knowing the user

- **User profile** - age, education, computer experience, domain knowledge, motivation, frequency of use, physical abilities.
- **Experience levels** - *novice* (needs guidance, few choices), *knowledgeable intermittent* (needs reminders and consistency), *expert* (needs speed, shortcuts, control).
- **Task characteristics** - frequency, criticality, complexity, sequence and constraints.
- **Environment** - lighting, noise, interruptions, mobility, shared devices.

Methods for gaining an understanding of users

Method	Description
Interviews and focus groups	Ask users about tasks, problems and needs; group discussion reveals shared opinions.
Questionnaires and surveys	Collect data from many users cheaply; quantitative analysis.
Observation / contextual inquiry	Watch users in their real environment; reveals what they *do* versus what they *say*.
Task analysis	Break tasks into steps and sub-goals; find inefficiencies.
Personas and scenarios	Fictional but data-based user archetypes and stories used to guide design.
Card sorting	Users group items to reveal how they expect information to be organised.
Usability testing	Users try prototypes while designers observe problems.

Laws that guide design

Law	Statement	Design use
Fitts's law	Time to reach a target grows with distance and shrinks with target size	Big, near buttons for frequent actions; screen edges and corners are easy to hit
Hick-Hyman law	Decision time increases with the number (and unpredictability) of choices	Limit choices, group and categorise menus
Miller's law	Working memory holds a limited number of chunks	Chunk information; avoid long unstructured lists of codes
Jakob's law	Users spend most time on other sites/apps and expect yours to work similarly	Follow conventions
Power law of practice	Performance improves with practice, with diminishing returns	Support learning curve; provide shortcuts

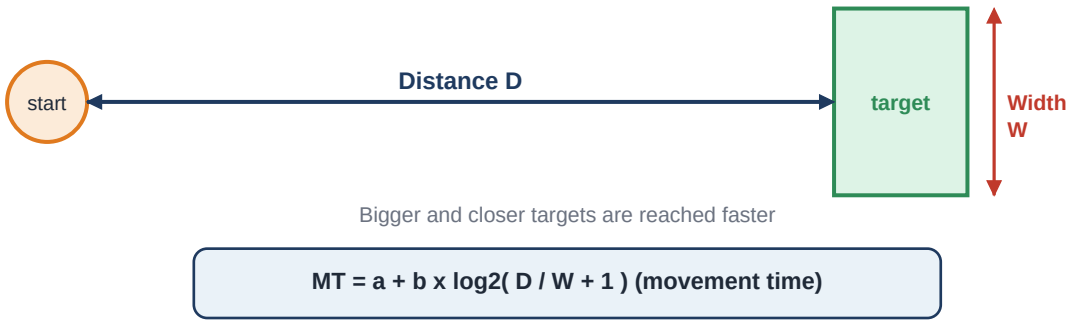


Figure 2.5: Fitts's law - movement time depends on distance D and target width W.

Other human considerations

- **Fatigue and ergonomics** - posture, glare, repetitive strain, screen distance, break needs.
- **Errors** - slips (right intention, wrong action) and mistakes (wrong intention). Design to prevent slips (spacing, confirmations) and mistakes (clear models, feedback).
- **Motivation and emotion** - frustration, anxiety and confidence affect performance; friendly, respectful wording helps.
- **Culture and language** - date formats, colours, reading direction, icons, humour and idioms differ.
- **Accessibility (WCAG - POUR)** - content must be *Perceivable*, *Operable*, *Understandable* and *Robust*. Examples: text alternatives, captions, keyboard access, sufficient contrast, resizable text, clear focus

indicators.

- **Performance versus preference** - what users say they like is not always what makes them fastest; use both measures and test.

2.5 Human Interaction Speeds

Human capabilities have time limits. Knowing the speed of reading, typing, pointing and reacting - and how quickly users expect the system to respond - is essential for setting design and performance targets.

Typical human speeds (approximate)

Activity	Typical value	Comment
Simple reaction time to a stimulus	150-300 ms	Longer with choice, fatigue or age
Perceptual / cognitive / motor cycle	~100 / ~70 / ~70 ms	Model Human Processor averages
Reading (paper, silent)	200-300 words per minute	Reading from a screen is typically slower, especially for poor contrast or small text
Speech	~120-180 words per minute	Voice input is faster than typing for many people but harder to edit
Typing (average / skilled typist)	~40 / 60-80+ words per minute	Strongly depends on practice
Mouse pointing to a target	Follows Fitts's law, around 1 s or less for typical targets	Small or distant targets take longer
Short-term memory decay	Seconds without rehearsal	Do not require carrying data between screens

System response time and user perception

The speed of the *system* must match the speed of the *human*. Widely used thresholds, from research by Miller and later Nielsen, are:

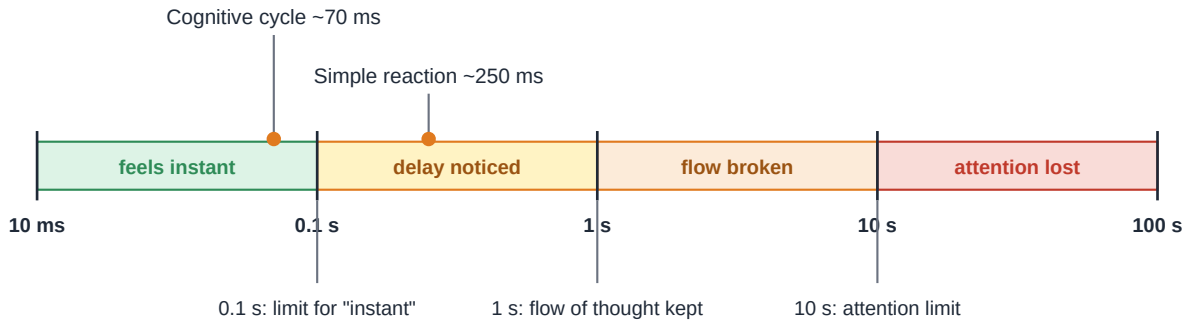


Figure 2.6: Response-time thresholds on a logarithmic scale.

Response time	User perception	Design guidance
Up to ~0.1 s	Feels instantaneous; direct cause and effect	Needed for typing echo, button press, dragging, animations - no special feedback needed
~0.1 to 1 s	Delay noticed, but flow of thought is maintained	Acceptable for most page loads and commands; no special indicator necessary

Response time	User perception	Design guidance
~1 to 10 s	User feels waiting; concentration begins to break	Show a busy indicator or skeleton; tell the user something is happening
Over ~10 s	Attention wanders; user may switch tasks or give up	Show percent-done progress, estimated time and allow cancel or background work

Impact of interaction speed on system performance and design

- **Productivity** - in high-volume data entry, saving even a second per transaction is significant over thousands of repetitions.
- **Errors and frustration** - slow or variable response times increase errors, repeated clicks and abandoned tasks.
- **Consistency of response** - predictable timing is often more important than raw speed.
- **Feedback techniques** - immediate acknowledgement (button state), busy cursor, progress bar, skeleton screens, optimistic updates.
- **Technical measures** - caching, compression, lazy loading, asynchronous processing, efficient queries.
- **Design for human speed** - avoid time-outs shorter than users need to read; avoid rapid flashing; allow users to control pacing.

2.6 Understanding Business Functions

An interface exists to support a **business function** - a set of tasks that help an organisation or a customer reach a goal (process an order, book a ticket, register a student). Designers must understand these functions in detail, otherwise even a beautifully designed screen will fail to support real work.

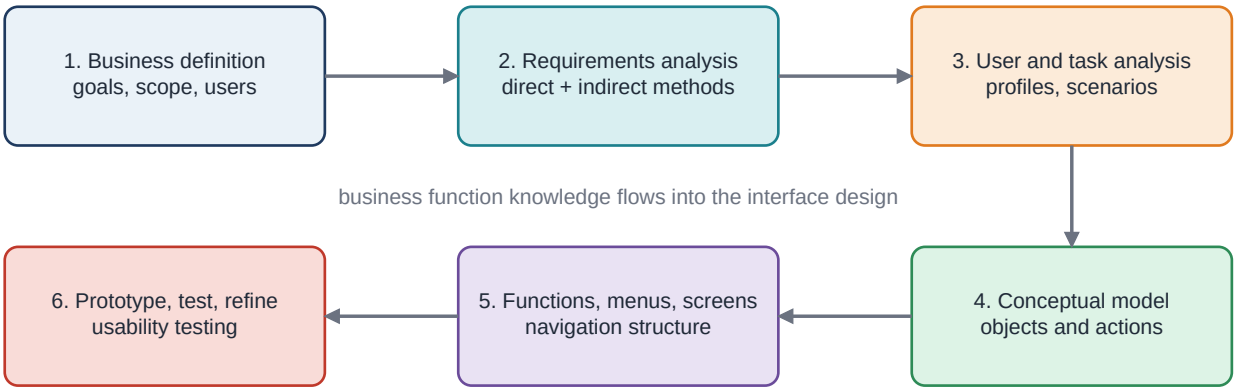


Figure 2.7: From business function to interface design.

Business definition and requirements analysis

The aim is to establish the **needs** of the users and the organisation, expressed as functional and non-functional requirements. Information may be gathered through direct or indirect contact with users.

Direct methods (contact with users)	Indirect methods (through others or data)
Face-to-face and telephone interviews	Intermediaries: managers, marketing, sales, support staff
Focus groups and facilitated sessions (JAD workshops)	Paper and electronic surveys and questionnaires
Observational field studies (visiting users)	Web-site or e-mail feedback, discussion forums
Prototyping and usability laboratory testing	Help-desk / support logs and complaint records

Direct methods (contact with users)	Indirect methods (through others or data)
Card sorting for information structure	Competitor and existing-system analysis; trade literature

Elements produced during analysis

- **User profiles** and **personas** - who will use the system.
- **Task analysis and workflows** - what steps users perform, in what order, with what information.
- **Use cases and scenarios** - stories of users achieving goals; used to test the design.
- **Functional requirements** - what the system must do (e.g. issue a receipt).
- **Non-functional requirements** - usability, performance, accessibility, security, platform, legal.
- **Usability goals** - measurable targets, e.g. "90% of new users complete registration within 3 minutes without help".

Developing a conceptual model

A **conceptual model** describes the objects users work with, the actions they perform on them, and how the objects relate. Designers identify nouns (objects) and verbs (actions) in the task descriptions and use them to define screens, menus and navigation. Interface objects should follow the terminology of the business, not the terminology of the programmer.

> **Example: online library system**
> **Objects:** member, book, loan, reservation, fine. **Actions:** search, borrow, return, renew, reserve, pay. **Screens derived:** search and results, book details, my loans, my account. **Requirements:** renewal in two clicks; overdue items highlighted; plain-language error messages; accessible to visually impaired users.

Determining basic business functions

From the requirements, the analyst lists the main functions, groups them logically (for example by task, user role or frequency) and orders them by importance and frequency. This grouping directly produces the **menu structure and navigation**: frequently used or critical functions should be visible and quick to reach, rarely used ones may be placed deeper.

Design standards and style guides

A **style guide** documents rules for the look and behaviour of an interface: terminology, colours, fonts, layout grids, control usage, message wording and interaction patterns. Examples: platform guidelines (Microsoft Fluent, Apple Human Interface Guidelines, Google Material Design) and organisation design systems. Benefits: consistency, faster design and development, reduced errors, easier maintenance and easier training.

System training and documentation

Even good interfaces need support. The documentation set should match user needs and be designed as carefully as the interface:

Type	Purpose
Quick-start / reference card	One-page essentials for new or occasional users
User guide / manual	Complete task-oriented explanation
Online help and search	Context-sensitive answers where the problem occurs
Tool-tips and inline hints	Short explanations at the point of use
Tutorials, onboarding tours, videos	Guided first-time learning
FAQs and support forums	Answers to common problems; community help

2.7 Self-Learning Guide (6 hours)

Task	What to do	Output
1. Human characteristics	Study perception, memory, attention, motor skill. Try a memory span test and a reaction-time test online.	Personal results and one-page summary of implications
2. User-centred design stages	Read ISO 9241-210 summary; map each stage to methods (interview, prototype, test).	Diagram of the UCD cycle with methods
3. Analyse real interfaces	Choose a banking app and a ticket-booking site; assess usability and accessibility (contrast, keyboard, screen-reader labels, text resize).	Checklist with findings
4. Interaction speeds	Measure page-load time and task time; compare with 0.1 / 1 / 10 s thresholds. Try Fitts's law: click targets of different sizes and record times.	Table of measurements and interpretation

Accessibility quick checklist

- Contrast ratio at least 4.5:1 for normal text (WCAG AA).
- All functions reachable by keyboard; visible focus.
- Images have text alternatives; video has captions.
- Colour is never the only carrier of meaning.
- Text can be enlarged to 200% without loss of content.
- Forms have labels and clear, specific error messages.

2.8 Team Activity Guide (3 hours)

Activity: analyse the design of an existing software or web application, identify human factors affecting usability, discuss the business requirements that shaped the interface, and present recommendations.

- **Step 1 (20 min) - Select** an application with a clear business purpose (online admission portal, food delivery, hospital appointment app).
- **Step 2 (30 min) - Business function analysis.** List goals, user groups, main tasks and the business rules visible in the interface.
- **Step 3 (50 min) - Human-factors analysis.** For each key screen, complete the table below.
- **Step 4 (30 min) - Recommendations.** Prioritise issues by severity and effort; sketch improved screens.
- **Step 5 (50 min) - Present** findings with screenshots, evidence (times, errors), and before/after mock-ups.

Human-factors analysis template

Screen / task	Human factor	Observation	Impact	Recommendation
Registration form	Memory	Asks to remember password rules after error	Errors, drop-off	Show rules beside field
Search results	Perception	Prices not aligned; poor grouping	Slow scanning	Align, group, whitespace
Checkout	Motor / speed	Small tap targets; 8 s payment wait with no feedback	Mis-taps, abandonment	Larger targets; progress indicator

Presentation outline (8-10 min)

1) Purpose and users of the application > 2) Business requirements identified > 3) Human-factor findings with evidence > 4) Recommendations with mock-ups > 5) Expected improvement and how you would test it.

2.9 Unit 2 Review Questions

- Explain the user-centred design process with a neat diagram. Why is it iterative?
- Discuss the obstacles and pitfalls in the interface development process and the common usability problems.
- Describe the Model Human Processor. What are its implications for interface design?
- Compare sensory, working and long-term memory. How should design accommodate their limits?
- Why are human characteristics important? Explain any five with their design consequences.
- State Fitts's law and Hick's law with applications.
- Explain human interaction speeds and the 0.1 s, 1 s and 10 s response-time limits.
- What is meant by understanding business functions? Explain direct and indirect methods of requirements gathering.
- What are style guides? What is the role of training and documentation in the design process?
- Choose an application and list its objects and actions to form a conceptual model.

Quick Revision: Key Terms

Term	One-line definition
HCI	Study and design of how people interact with computers.
Usability	Effectiveness, efficiency and satisfaction in a context of use.
GUI / WIMP	Graphical interface based on Windows, Icons, Menus, Pointer.
Direct manipulation	Acting on visible objects with immediate, reversible, incremental feedback.
Metaphor	Familiar real-world concept used to explain a computer function (desktop, trash).
Affordance	Perceived cue for how an object can be used.
Gulf of execution / evaluation	Gap between intention and available actions / between system state and its perception.
UCD	Design process centred on users' needs, with iterative evaluation.
Mental model	User's internal understanding of how a system works.
Working memory	Small, short-lived store used for current tasks.
Fitts's law	Movement time depends on distance to and size of target.
Persona	Research-based fictional user archetype used to guide design.
Style guide	Documented standards for the appearance and behaviour of an interface.